

CONFIDENCE INTERVALS

master recipe • also used in Units 4 and 5

ZONE A · THE FORMULAS

statistic ±

(critical value)(standard error)

margin of error

ME = z^* · SE (proportions)

ME = t^* · SE (means)

width $\propto \frac{1}{\sqrt{n}}$ $\Rightarrow$ halve the margin $\Rightarrow$ quadruple n

THE FOUR-STEP FRAME

1	STATE	Name the parameter in context.
2	PLAN	Name the interval; check conditions.
3	DO	Formula, substitutions, and interval.
4	CONCLUDE	Interpret the interval in context.

COMMON z^* VALUES

confidence	z^*
90%	1.645
95%	1.960
99%	2.576

Get t^* from the table or calculator using **df**.

ZONE B · CONDITIONS — ALWAYS IN THIS ORDER

1 · RANDOM

Random sample or randomized experiment.

2 · 10%

Sampling without replacement:
 $n < 0.10N$.

3 · NORMAL / LARGE COUNTS

proportions: $n\hat{p} \geq 10$, $n(1 - \hat{p}) \geq 10$
means: normal population, $n \geq 30$, or no strong skew/outliers

ZONE C · SAY IT IN WORDS

“We are ____% confident that the interval from ____ to ____ captures the true ____.”

Confidence level: If we took many samples and built an interval each time, about ____% of those intervals would capture the true ____.

ZONE D · WATCH OUT

THE INTERVAL IS ABOUT THIS SAMPLE. THE LEVEL IS ABOUT THE METHOD.

Never say the fixed parameter has a 95% chance of being inside the interval. After the interval is calculated, it either captures the parameter or it does not.